SEMESTER PROJECT PROPOSAL

Project Title

Phishing URL Detection System Using FA and Regex

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Course: Theory of Automata

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1. Introduction

Cybersecurity threats have increased significantly with the rapid growth of internet technologies and online services. One of the most common cyber attacks today is phishing, where attackers create fake or malicious websites to steal sensitive user information such as passwords, banking credentials, and personal data.

Phishing websites often imitate trusted websites by using suspicious keywords, misleading domain names, and deceptive URL structures. Detecting such URLs manually is difficult for users, making automated phishing detection systems highly important.

This project aims to develop a phishing URL detection system using concepts from Theory of Automata. The proposed system will analyze URLs using Regular Expressions and Deterministic Finite Automata (DFA) to identify suspicious patterns and classify URLs as safe or malicious.

The project combines cybersecurity concepts with automata theory to provide a practical and educational implementation of pattern recognition techniques.

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2. Problem Statement

Phishing websites are becoming increasingly sophisticated and difficult to identify manually. Many users unknowingly visit malicious websites and provide sensitive information to attackers.

There is a need for a lightweight automated system that can analyze URLs and identify suspicious patterns before users access potentially dangerous websites.

This project aims to solve this problem by implementing a phishing URL detection system using finite automata and pattern matching techniques.

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3. Objectives

The main objectives of this project are:

1. To understand and apply Theory of Automata concepts in cybersecurity.
2. To analyze URL structures and identify suspicious patterns.
3. To implement Regular Expressions for URL pattern matching.
4. To design DFA-based transitions for phishing detection.
5. To classify URLs as safe or suspicious.
6. To generate alerts for potential phishing attempts.

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4. Scope of the Project

The project will focus on detecting suspicious URLs based on predefined phishing patterns such as:

- Suspicious keywords
- Fake domain structures
- Unusual symbols
- Dangerous extensions

The project will not include:

- Real-time browser integration
- Machine learning algorithms
- Advanced web crawling
- Enterprise-scale threat intelligence systems

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5. Theory of Automata Concepts Used

5.1 Finite Automata

The project will use DFA states to classify URLs.

Example states:

- q0 = Start State
- q1 = Suspicious Keyword State
- q2 = Suspicious Domain State
- q3 = Phishing Alert State

5.2 Regular Expressions

Regular Expressions will be used to detect phishing patterns.

Examples:

- login
- verify
- secure
- banking
- .xyz
- .ru

5.3 State Transitions

The DFA will move between states based on detected patterns.

Example: q0 → q1 when suspicious keyword is detected

q1 → q2 when suspicious domain extension is found

q2 → q3 when multiple suspicious patterns exist

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6. Methodology

The project methodology includes:

- Studying phishing URL patterns
- Designing DFA diagrams
- Implementing URL analysis using Python
- Applying Regular Expressions
- Detecting suspicious patterns

- Generating phishing alerts
- Testing using sample URLs

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7. Tools and Technologies

Tool/Technology	Purpose
Python/ C++	Main programming languages
Regular Expressions	Pattern matching
VS Code	Code editor
JFLAP / Graphviz	DFA visualization
Text Files	URL datasets

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8. Expected Output

The system will:

- Accept URLs as input
- Analyze suspicious patterns
- Detect phishing indicators
- Display security alerts

Example output:

======================== PHISHING URL DETECTOR ========================

Checking URL: http://paypal-secure-login.xyz

Suspicious Keyword Detected

Suspicious Domain Detected

ALERT: Possible phishing website identified

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9. Significance of the Project

This project demonstrates how automata theory can be applied in practical cybersecurity systems. It introduces concepts such as:

- Pattern recognition
- String matching

- State-based analysis
- Threat detection

The project also provides exposure to real-world cybersecurity problems related to phishing attacks and malicious websites.

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10. Future Enhancements

Future improvements may include:

- Browser extension integration
- Real-time website scanning
- Machine learning-based classification
- Online blacklist databases
- GUI implementation

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11. Conclusion

The proposed phishing URL detection system aims to identify suspicious URLs using finite automata and regular expressions. By applying Theory of Automata concepts in cybersecurity, the project provides a practical implementation of phishing detection techniques.

The project will enhance understanding of automata theory while also introducing important concepts related to web security and cyber threat detection.

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12. References

1. Introduction to Automata Theory, Languages, and Computation – Hopcroft & Ullman
2. Python Official Documentation
3. Regular Expression Documentation
4. Cybersecurity and Phishing Detection Resources